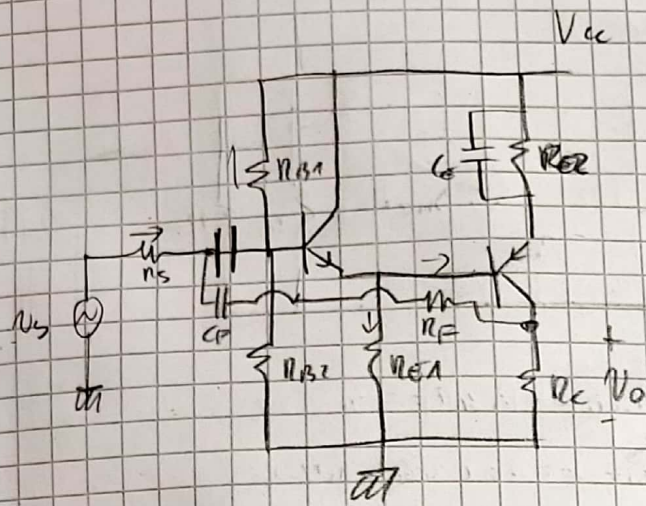




1) Value of R_2 such that $V_o = 10.5V$

Therm on input

$$V_B = \frac{V_{CC} R_2}{R_1 + R_2} = 16.15V$$

$$R_B = R_1 // R_2 = 48.71k\Omega$$

$$I_{C2} = \frac{V_o}{R_C} = 1.875mA \quad I_{B2} = 0.0125mA$$

$$V_{B1} - R_B I_{B1} - V_{BE1} - R_{E1} (I_{E1} - I_{B2}) = 0$$

$$V_{B1} - R_B I_{B1} - V_{BE1} - R_{E1} I_{B1} (1 + \beta_{F1}) + R_{E1} I_{B2} = 0$$

$$I_{B1} = \frac{V_{B1} - V_{BE1} + R_{E1} I_{B2}}{R_B + R_{E1} (1 + \beta_{F1})} = 0.01635mA$$

$$I_{C1} = 3.266mA$$

$$V_{CE1} = V_{CC} - R_{C1} (I_{C1} - I_{B2}) = 11.25V$$

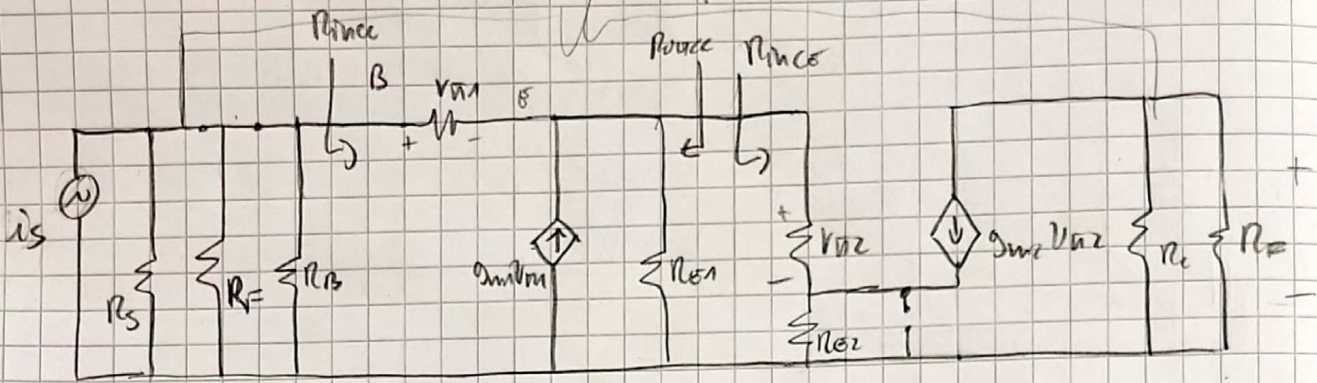
$$R_{E2} = \frac{V_{CC} - V_{CE1} - V_{BE2}}{I_{C2}} = 5.59k\Omega$$

$$V_{E2} = V_{CC} - V_{CE1} + V_{BE2} = 13.45V - 0.7V = 12.75V$$

$$V_{CE2} = 2.95V$$

$$V_{CE2} = 10.5V = V_o$$

3) $A_v = V_o/V_s$, current mirroring $\rightarrow$ Norton $R_{D1} = R_{D2} = R_E = \beta = -1/g_{m2}$



$$g_{m1} = 125,6 \text{ mS}^{-1}$$

$$g_{m2} = 72,1 \text{ mS}^{-1}$$

$$R_{out1} = R_E \parallel R_F = 3,56 \text{ k}\Omega$$

$$R_{in1} = r_{\pi 2} = 2,08 \text{ k}\Omega$$

$$r_{\pi 1} = 1,6 \text{ k}\Omega$$

$$r_{\pi 2} = 2,08 \text{ k}\Omega$$

$$R_x = R_S \parallel R_E \parallel R_F = 0,892 \text{ k}\Omega$$

$$R_{out2} = \frac{r_{\pi 1} + R_x}{1 + \beta_{01}} = 123,5 \Omega$$

$$R_{in2} = r_{\pi 1} + (1 + \beta_{01})(R_{out1} \parallel R_{in1}) = 274,76 \text{ k}\Omega$$

$$A_{vol} = \frac{V_o}{i_s} = A_{v1} \cdot A_{v2} \cdot \frac{V_i}{i_s}$$

$$\frac{V_i}{i_s} = R_{in}^{ol} = 0,89 \text{ k}\Omega$$

$$A_{v1} = \frac{(1 + \beta_{01})(R_{E1} \parallel r_{\pi 2})}{r_{\pi 1} + (1 + \beta_{01})(R_{E1} \parallel r_{\pi 2})} = 0,9941$$

$$A_{v2} = -g_{m2}(R_E \parallel R_F) = -258,8$$

$$1 + T = 23,897$$

$$A_{vol} = 0,89 \cdot 0,9941 \cdot (-258,8) = -228,973 \text{ k}\Omega$$

$$A_v^F = \frac{A_v^a}{1 + \beta A_{v1}} = -0,58 \text{ k}\Omega$$

$$A_v = A_{vol} \frac{1}{R_S} = -9,58$$

4) $R_{in}^F = \frac{R_{in}^{ol}}{1 + T} = 37,25 \Omega$ $R_{in} = \frac{1}{\frac{1}{R_{in}^F} - \frac{1}{R_S}} = 38,7 \Omega$

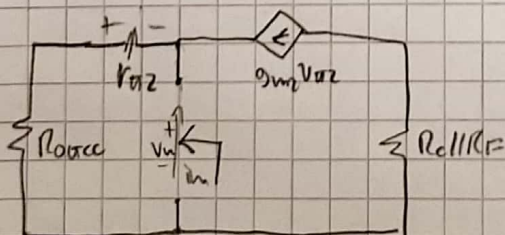
5) determine f_L given by C_E applying feedback!

$$R_E^{sc} = R_{E2} \parallel R_F$$

$$i_m = -\frac{v_{\pi 2}}{r_{\pi 2}} - g_{m2} v_{\pi 2} = -\frac{v_{\pi 2}}{r_{\pi 2}} (1 + \beta_{02})$$

$$v_m = -\frac{v_{\pi 2}}{r_{\pi 2}} (R_{out2} + r_{\pi 2})$$

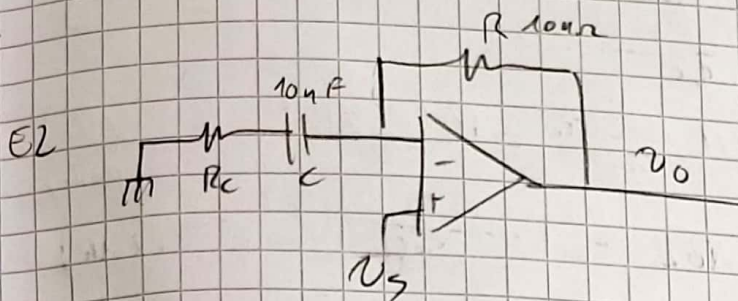
$$R_E^{sc} = 14,56 \Omega$$



$$R_F = \frac{v_m}{i_m} = \frac{R_{out2} + r_{\pi 2}}{1 + \beta_{02}} = 14,6 \Omega$$

$$f_L = \frac{1}{20 C_E R_E^{sc}} = 10,9$$

$$f_L = 457 \text{ Hz}$$



$$A(s) = \frac{A_{d0}}{1 + s/\omega_0}$$

$$\omega_0 = 500 \text{ rad/s}$$

$$R_c = 0.2 \Omega$$

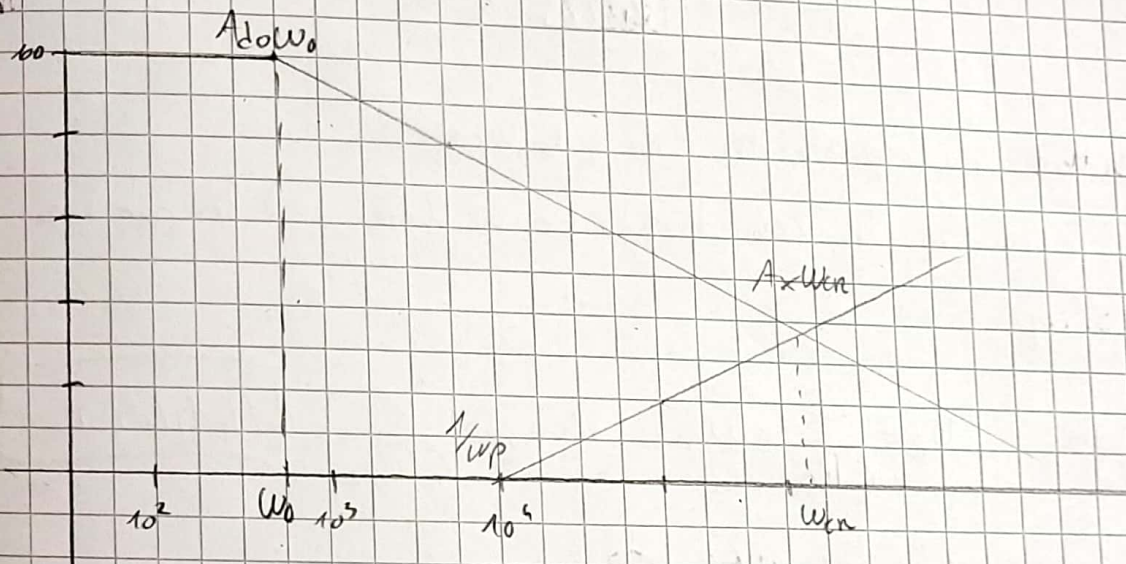
1) ideal circuit gain

$$A_v(s) = 1 + \frac{Z_2}{Z_1} = 1 + \frac{R}{1 + sCR_c} = 1 + \frac{sCR}{1 + sCR_c} = \frac{1 + sC(R + R_c)}{1 + sCR_c}$$

$$\omega \rightarrow \infty \rightarrow$$

$$A_v \omega = 1 + \frac{R}{R_c}$$

2) phase margin:



$$T = \frac{v_o}{v_i} = A_d(s) \cdot \frac{1 + sCR_c}{1 + sC(R_c + R)} = A_d(s) \cdot \frac{1 + sCR_c}{1 + sC(R + R_c)} \quad \begin{matrix} R_c = 0 \\ \hline = A_d(s) \cdot \frac{1}{1 + sCR} \end{matrix}$$

$$B(s) = \frac{1}{1 + sCR} \quad \omega_p = \frac{1}{CR} = 10^4 \text{ rad/s} \quad \Delta m = \Delta m = 40 \text{ dB} \rightarrow \text{amph!!}$$

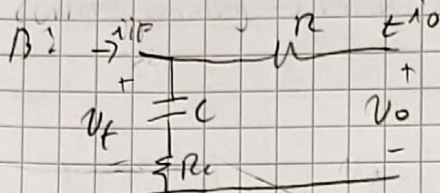
$$\begin{cases} A_{ol} \omega_0 = A \times \omega_{cn} \\ \frac{1}{\omega_p} = \frac{A}{\omega_{cn}} \end{cases}$$

$$\omega_{cn} = \sqrt{A_{ol} \omega_0 \omega_p} = \sqrt{7,07 \cdot 10^5} \text{ rad/s}$$

$$\phi_m = 180 - 2 \arctan\left(\frac{\omega_{cn}}{\omega_p}\right) - 2 \arctan\left(\frac{\omega_{cn}}{\omega_p}\right) = 0,85^\circ \text{ low!}$$

3) BD (feedback trans)

Voltage mixing

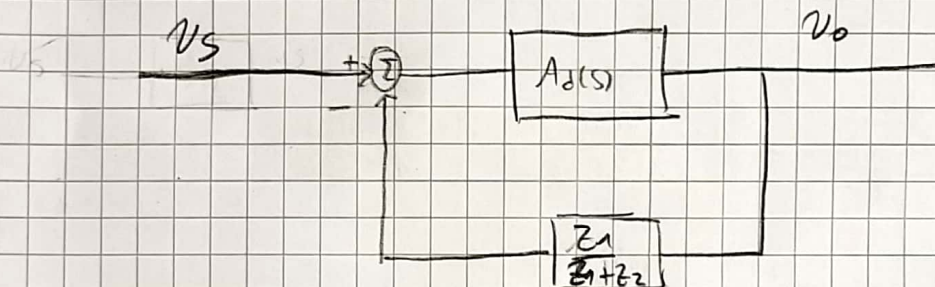


$$\begin{cases} v_F = \beta v_o + R_i i_F \\ i_{OF} = v_o / R_o + \delta i_F \end{cases}$$

$$\beta = \frac{v_F}{v_o} \Big|_{i_F=0} = \frac{1 + sCR_c}{sC} = \frac{1 + sCR_c}{1 + sC(R_c + R)} = \frac{z_1}{z_1 + z_2}$$

$$R_{oi} = \frac{v_o}{i_F} \Big|_{v_o=0} = \frac{1}{sC} + R + R_c = \frac{1 + sC(R + R_c)}{sC} = \frac{z_1 + z_2}{sC}$$

$$R_{if} = \frac{v_F}{i_F} \Big|_{v_o=0} = R \parallel \frac{1 + sCR_c}{sC} = \frac{R(1 + sCR_c)}{sC} = \frac{R(1 + sCR_c)}{1 + sC(R + R_c)} = \frac{z_1}{z_1 + z_2}$$



4) max R_c such that freq resp stay the same up to ω_{cn}

R_c introduces a zero so the zero has to be at ω_{cn} and the pole has to stay there at ω_p

$$\omega_{cn} = 7,07 \cdot 10^5 \text{ rad/s} \quad \omega_z = \frac{1}{CR_c} = \omega_{cn} \quad R_{c,max} = \frac{1}{C\omega_{cn}} = 141,45 \Omega$$

bigger and the zero comes to the left $\smile$

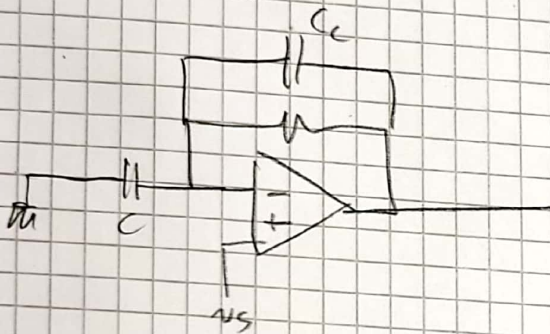
5) estimation of the phase margin. it should be at least 45°

for precision, calculate new ω_{pc} $\omega_{pc} = \frac{1}{C(R+R_c)} = 9860 \text{ rad/s}$

$$\phi_{mc} = 180 - \arctan\left(\frac{\omega_{en}}{\omega_0}\right) - \arctan\left(\frac{\omega_{en}}{\omega_{pc}}\right) + \arctan\left(\frac{\omega_{en}}{\omega_z}\right) = \boxed{45.83^\circ}$$

6) equivalent compensation strategy:

Instead of placing a series R_c i put a parallel C_c



$$T = \frac{u_v}{u_e} = A_d(s) \cdot \frac{1/sC}{\frac{1}{sC} + \left(\frac{1}{sC} + R\right)}$$

$$A_d(s) \cdot \frac{1/sC}{\frac{1}{sC} + \frac{R}{1+sCR}} = \frac{1/sC}{\frac{1}{sC} + \frac{R}{1+sCR}}$$

$$= \frac{1/sC}{\frac{1+sCR+sCR}{sC(1+sCR)}} = \frac{1+sCR}{1+s(C+C_c)R}$$

to size it $C_c = \frac{1}{\omega_{en}R} = \boxed{14.14 \text{ nF}}$

14.125 nF (can simplify su block diagram)